STAT 253
Probability & Statistics

January 31, 2024

Course Objective

To develop a basic understanding of probability theory as a tool for analysis of random events arising from various fields of science.

Course Syllabus

- Introduction to Probability Theory
- Random Variables and Probability Distributions
- Discrete and Continuous Distributions
- Moments and Moment Generating Functions
- Probability Theorems (e.g Central Limit Theorem, etc.) and their Applications

References/Reading List

The following materials will be used as references and reading list for this course;

1 F.M. Dekking, C. Kraaikamp H.P. Lopuhaä and L.E. Meester (2005). A Modern Introduction to Probability and Statistics: Understanding How and Why, Springer-Verlag, Berlin.

2 Ghahramani, S. (2018). Fundamentals of Probability: With Stochastic Processes, Chapman and Hall/CRC, New York.

2 Ross, S. M. (2014). A first course in probability, Pearson, New York.

3 Grimmett, G., & Stirzaker, D. (2020). Probability and Random Processes, Oxford University Press, England.

4 Sheldon M Ross(2004), Introduction to Probability & Statistics, for Engineers & Scientists, 5th Edition, Academic Press, New York.

Read from these books and more

Intended Outcome

In the first part of this course we will look at random experiments, determination of measure of probability, basic laws of probability involving compound events, etc.

After studying this unit students should be able to

- Understand probability measure and how it is determined
- Know some basic terms, rules or theorems of probability
- Apply the rules or theorems of probability to compute probabilities of events

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I-1: Set Theory

Recall:

Consider a given set Ω and the power set, $\mathcal{P}(\Omega)$, consisting of all subsets of Ω. There are some set-theoretic operations we can perform on subsets of Ω.

- The set: $\emptyset_\Omega = \emptyset = \{x \in \Omega; x \neq x\}$ is the **empty subset** of Ω

Union: If A and B are subsets of Ω, we define there union $A \cup B$ to be the set of elements that are in either A or B. That is

$$A \cup B = \{x \in \Omega \mid x \in A \text{ or } x \in B\}$$

NB:

$$A \cup B = B \cup A$$
$$A \cup (B \cup C) = (A \cup B) \cup C$$

Try some examples

What is the power set, $\mathcal{P}(\Omega)$ of the set $\Omega = \{a, b\}$

Set Theory

Intersection: If A and B are subsets of Ω, we define their intersection $A \cap B$ to be the set of all elements which belong to both A and B. Thus;

$$A \cap B = \{x \in \Omega \mid x \in A \text{ and } x \in B\}$$

NB:

$$A \cap B = B \cap A \text{ Also; } A \cap (B \cap C) = (A \cap B) \cap C$$

Remark: The empty set $\emptyset$ and Ω play special roles

$$A \cup \emptyset = A \qquad A \cap \emptyset = \emptyset$$
$$A \cup \Omega = \Omega \qquad A \cap \Omega = A$$

Try some examples

Set Theory

Compliment: If A is a subset of Ω, we define the compliment, $\bar{A}$ (A^c) of A (relative to Ω) as the set of elements not in A. Thus;

$$\bar{A} = \{x \in \Omega \mid x \notin A\}$$

Remark:

$$\bar{\emptyset} = \Omega, \qquad \bar{X} = \emptyset \quad (X^c = \emptyset)$$
$$A \cap A^c = \emptyset, \qquad (A^c)^c = \overline{(\bar{A})} = A$$

We will also discuss ; **Equality of Sets, Difference of Sets, Difference & Complements, and Partition of Sets.**

We try some examples in class

Set Theory: De Morgan's Laws

Let A and B be two subsets of Ω.

- First Law: $(A \cup B)^c = A^c \cap B^c$
- Second Law: $(A \cap B)^c = A^c \cup B^c$

In general consider $A_1, A_2, \cdots, A_n$ as subsets of Ω. Then

$$1.\ \left(\bigcup_{i=1}^{n} A_i\right)^c = \bigcap_{i=1}^{n} A_i^c \quad ; \quad \left(\bigcup_{i=1}^{\infty} A_i\right)^c = \bigcap_{i=1}^{\infty} A_i^c$$

$$2.\ \left(\bigcap_{i=1}^{n} A_i\right)^c = \bigcup_{i=1}^{n} A_i^c \quad ; \quad \left(\bigcap_{i=1}^{\infty} A_i\right)^c = \bigcup_{i=1}^{\infty} A_i^c$$

Augustus De Morgan (1806 - 1871) was a British Mathematician

I-2: Illustration of I-1

If an experiment consists of flipping a coin, then;

$$\Omega = \{H, T\} = \{H\} \cup \{T\} \tag{1}$$

Consider $\{H\} \subset \Omega$ as a subset of Ω. Then

$$\{H\}^c = \{T\} \text{ and } \{T\}^c = \{H\} \tag{2}$$

Replacing by $\{T\}$ and $\{H\}$ in equation (1) with (2) we have;

$$\Omega = \{T\}^c \cup \{H\}^c = \{H\}^c \cup \{T\}^c$$

From De Morgan's second Law

$$\Omega = \{H\}^c \cup \{T\}^c = (\{H\} \cap \{T\})^c$$

Illustration Continued

But

$$\{H\} \cap \{T\} = \emptyset$$

Therefore

$$\Omega = \emptyset^c \Leftrightarrow \Omega^c = \emptyset \tag{3}$$

Next we demonstrate the relationship between these set jargons and probability terms.

Revise/Exercise

Suppose you have 3 circuit boards, A, B, C, what is the mathematical expression or set notation for the following;

(a) Only A occurs
(b) Both A and B
(c) At least two events occur
(d) At least one event occur
(e) One and only event occurs
(f) All 3 events occur
(g) Not more than 2 events occur
(h) Exactly 2 events occur

Try and work out more examples

I-3: Random Experiment, Sample Spaces and Events

Definition (Random Experiment)

A random experiment is any experiment for which one cannot predict the outcome(s) with certainty. Example: Tossing a coin, flipping a six - sided die.

The **sample space**, usually denoted by Ω or S for a random experiment is the set of all possible outcomes (these possible outcomes are all predictable in advance).
Example 1: Tossing a coin

$$\Omega = \{H, T\} \tag{4}$$

Example 2: Rolling a die ; $\Omega = \{1, 2, 3, 4, 5, 6\}$. Its size, denoted by $\#\Omega$ or $n(\Omega)$ is 6, which is finite.

Exx2: In an experiment we ask the next person we meet on the street in which month her birthday falls. What is the sample space?

I-3 Continued: Examples

- **Example 3**: Tossing a fair coin twice. What is the sample space?
- **Exx** What is the sample space associated with the following experiments
  - Select a letter from the word SAMPLE
  - Outcome of a football match
- **Example 4**: Observing the lifetime (in hours) of a light bulb is a random experiment with sample space $\Omega = [0, \infty)$. The sample space is infinite.
- **Remark**: Given any random experiment, its sample space, Ω can either be finite or infinite.

Sample Space and Events

Definition (Events)

The outcomes of a a random experiment are called Events (subsets of a sample space).

- **Example**: In equation (4), $\{H\}$ is an event.
- **Exx2**, What is the outcome(s) that correspond to a month with 31 days? This is the event.
- In the roll a die in Example 2. Find the events that correspond to the phrases
  - an even number is rolled
  - a number greater than two is rolled
  - a number one and six is rolled

Can you think of other experiments, their sample spaces and some events?

I-3-1: Special Events in a Random Experiment

For any sample space, Ω of a random experiment, we have

Definition (Subset)

An event A is said to be a subset of the event B if whenever A occurs, B also occurs.

Definition (Equality)

Events A and B are said to be equal if the occurrence of A implies the occurrence of B and vice versa, i.e

$$A = B \iff \begin{cases} A \subseteq B \\ B \subseteq A \end{cases}$$

I-3-1: Continued

Definition (The Certain Event)

An event is called certain if its occurrence is inevitable. Thus the sample space, Ω is a certain event.

Definition (The Impossible Event)

An event is called impossible if there is certainty in its non-occurrence. Therefore, the empty set $\emptyset$, which is Ω^c, is an impossible event.

Definition (Intersection)

An event is called the intersection of two events A and B if it occurs only whenever A and B occur simultaneously. It is denoted by $A \cap B$

I-3-1 Continued

Definition (Union)

An event is called the union of two events A and B if it occurs whenever at least one of them occurs. It is denoted by $A \cup B$

Definition (Compliment)

An event is called the complement of the event A if it only occurs whenever A does not occur. The complement of A is denoted by A^c.

Definition (Difference)

An event is called the difference of two events A and B if it occurs whenever A occurs but B does not. The difference of the events A and B is denoted by

$$A - B = A \cap B^c$$

I-3-1 Continued

Definition (Mutually Exclusive)

If the joint occurrence of two events A and B is impossible, we say that A and B are mutually exclusive.

A and B are mutually exclusive if $A \cap B = \emptyset$. A set of events $\{A_1, A_2, \cdots\}$ is called mutually exclusive if the joint occurrence of any two of them is impossible, that is, if $\forall i \neq j$, $A_i \cap A_j = \emptyset$. Thus $\{A_1, A_2, \cdots\}$ is mutually exclusive if and only if every pair of them is mutually exclusive.

Definition (Mutually Exhaustive)

Two or more events are said to be **exhaustive** if their union equals the equals the sample space, Ω.

Exercise

We toss a coin 3 times. We give the sample size as

$$\Omega = \{HHH, THH, HTH, HHT, TTH, THT, HTT, TTT\}$$

where T stands for tails and H stands for heads.

1. Write down the set of outcomes corresponding to each of the following events:
   A: "we throw tails exactly two times"
   B: "we throw tails at least two times"
   C: "tails did not appear before a head appeared"
   D: "the first throw results in tails"

2. Write down the set of outcomes corresponding to each of the following events:
   A^c, $A \cup (C \cap D)$, and $A \cap D^c$

Try it!

I-4-1:Probability Continued

Definition (Probability)

Probability measures the likelihood of an event occurring.

We outline 3 of the possible ways probability can be interpreted;

Classical Definition of Probability

If a random experiment can result in $n(S)$ mutually exclusive and equally likely outcomes and if $n(A)$ of these outcomes result in the occurrence of the event A, the probability of A is defined as

$$\mathbb{P}(A) = \frac{\text{number of succesful outcomes}}{\text{number of possible outcomes}} = \frac{n(A)}{n(S)} \tag{5}$$

This is based on the assumption that all the possible outcomes of the experiment are equally likely. E.g Toss of a fair coin

Example

A fair die is tossed once. What is the probability of ;

1. Observing a 3
2. Observing an odd number
3. Observing any number

what can we say about (3)

Probability Continued

Relative Frequency Definition

If a random experiment is repeated a large number of times, say n times, under identical conditions and if an event A is observed to occur $n(A)$ times, then the probability of the event A is defined as

$$\mathbb{P}(A) = \lim_{n \to \infty} \frac{n(A)}{n} \tag{6}$$

However, in practise, we can make a moderate number of trial E.g 260 bolts are examined as they are produced. Five of them are found to be defective. On the basis of this information, estimate the probability that a bolt will be defective. **Ans** $= \frac{5}{260}$

Example

The frequency table shows the age distribution of 20 students in a class.

Age in years	9	10	11	12
Frequency	2	8	6	4

What is the probability that a student selected at random is

- 10 years old
- 11 years old
- less than 11 years old

Try them as a form of revision

Subjective Definition

Another type of probability is the subjective estimate, based on a person's experience.

Example Say a geological engineer examines extensive geological information on a particular property. He chooses the best site to drill an oil well, and he states that on the basis of his previous experience he estimates that the probability the well will be successful is 30%. (Another experienced geological engineer using the same information might well come to a different estimate.)

In 1933 **Andrew Kolmogorov** proposed the Axiomatic definition of probability which is still dominant and will be discussed next.

Andrew Kolmogorov (1903 - 1987) was a Russian - ex Soviet Mathematician

I-4: Axiomatic Definition of Probability

Consider an experiment whose sample space is Ω. A real-valued function $\mathbb{P}$ on the space of all events of the experiment is called a probability measure if;

- **Axiom 1** : For any event, A of the sample space, Ω

$$0 \leq \mathbb{P}(A) \leq 1 \tag{7}$$

  we call $\mathbb{P}(A)$ the probability of the event A

- **Axiom 2**

$$\mathbb{P}(\Omega) = 1$$

Kolmogorov's axioms of probability

I-4: Continued

- **Axiom 3**: For any sequence of events $A_1, A_2, \cdots$, that are mutually exclusive, that is events for which $A_i \cap A_j \neq \emptyset$ when $i \neq j$ then

$$\mathbb{P}\left(\bigcup_{k=1}^{\infty} A_k\right) = \sum_{k=1}^{\infty} \mathbb{P}(A_k) \tag{8}$$

For 2 events A_1 and A_2 we have

$$\mathbb{P}(A_1 \cup A_2) = \mathbb{P}(A_1) + \mathbb{P}(A_2) \tag{9}$$

Example 1

Let's consider the rolling die example

If a die is rolled and we suppose that all six sides are equally likely to appear, what is the probability of rolling an even number.

$$\Omega = \{1, 2, 3, 4, 5, 6\}$$

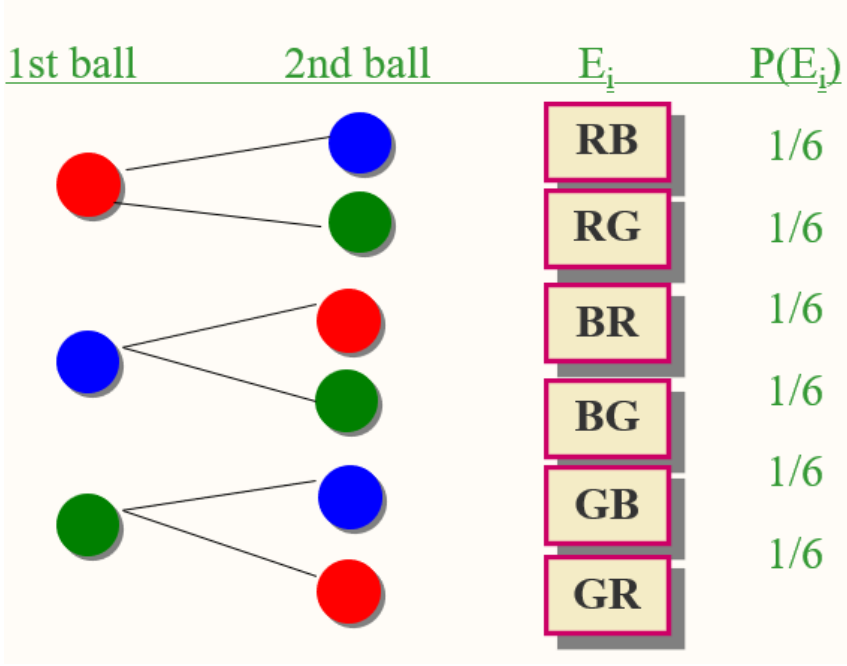
$$P(1) = P(2) = P(3) = P(4) = P(5) = P(6) = \frac{1}{6} > 0$$

The event A of rolling an even number is $A = \{2, 4, 6\}$. From Axiom 3, it would thus follow that the probability of rolling an even number would equal

$$\mathbb{P}(A) = \mathbb{P}(\{2, 4, 6\}) = P(2) + P(4) + P(6) = \frac{1}{2}$$

Example 2

A bowl contains 3 balls, one red, one blue and one green. A child selects two balls at random. What is the probability that at least one is red.



$$\mathbb{P}(\text{at least 1 red}) = \mathbb{P}(RB) + \mathbb{P}(BR) + \mathbb{P}(RG) + \mathbb{P}(GR)$$
$$= \frac{1}{6} + \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = \frac{4}{6} = \frac{2}{3}$$

I-4-2: Properties of Probability

Theorem

Suppose that $\mathbb{P}$ is a probability measure. Then it satisfies the following properties.

1. The probability of the empty set is 0 i.e $\mathbb{P}(\emptyset) = 0$
2. For any event A, $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$
3. If $A \subseteq B$ then $\mathbb{P}(A) \leq \mathbb{P}(B)$
4. For any events A and B, $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$

We will show proofs of properties $(1) - (4)$

Property (4) is the **Addition Law of Probability**.

I-4-2 Continued

For three events A_1, A_2 and A_3 we have

$$\mathbb{P}(A_1 \cup A_2 \cup A_3) = \mathbb{P}(A_1) + \mathbb{P}(A_2) + \mathbb{P}(A_3) - \mathbb{P}(A_1 \cap A_2) - \mathbb{P}(A_1 \cap A_3) - \mathbb{P}(A_2 \cap A_3) + \mathbb{P}(A_1 \cap A_2 \cap A_3) \tag{10}$$

Theorem

$\mathbb{P}(A) = \mathbb{P}(A \cap B) + \mathbb{P}(A \cap B^c)$ **Proof:**

$$A = A \cap \Omega = A \cap (B \cup B^c) = (A \cap B) \cup (A \cap B^c)$$

But, $(A \cap B)$ *and* $(A \cap B^c)$ *are mutually exclusive (verify) Hence,*

$$\mathbb{P}(A) = \mathbb{P}(A \cap B) + \mathbb{P}(A \cap B^c) \tag{11}$$

Example 1

Suppose that we toss two coins, and we assume that each of the four outcomes in the sample space

$$\Omega = \{HH, HT, TH, TT\}$$

is equally likely and hence has probability $\frac{1}{4}$. What is the probability that either the first or the second coin falls on a head?

Let E be the event that the first coin falls on a head, and F, the event that the second coin falls on a heads. Then

$$E = \{HH, HT\} \text{ and } F = \{HH, TH\}$$

Then the probability that either the first or the second coin falls heads, is given by

Solution to Example 1 Cont'd

$$\begin{aligned}\mathbb{P}(E \cup F) &= \mathbb{P}(E) + \mathbb{P}(F) - \mathbb{P}(E \cap F) \\ &= \frac{1}{2} + \frac{1}{2} - \frac{1}{4} = \frac{3}{4}\end{aligned}$$

Example 2

Find more examples in tutorial set 1

Example 3

A random experiment can result in one of the outcomes $\{a, b, c, d\}$ with probabilities 0.1, 0.3, 0.5, and 0.1, respectively. Let A denote the event $\{a, b\}$, B the event $\{b, c, d\}$, and C the event $\{d\}$. Find $\mathbb{P}(A)$, $\mathbb{P}(B)$, $\mathbb{P}(C)$ and their complements

Solution

$$\mathbb{P}(A) = 0.1 + 0.3 = 0.4, \quad \mathbb{P}(B) = 0.3 + 0.5 + 0.1 = 0.9, \quad \mathbb{P}(C) = 0.1$$

$$\mathbb{P}(A^c) = 0.6, \quad \mathbb{P}(B^c) = 0.1, \quad \mathbb{P}(C^c) = 0.9$$

Also find $\mathbb{P}(A \cup B)$, $\mathbb{P}(A \cap B)$ and $\mathbb{P}(A \cap C)$

$A \cup B = \{a, b, c, d\}$ $\mathbb{P}(A \cup B) = 0.1 + 0.3 + 0.5 + 0.1 = 1.0$

Because $A \cap B = \{b\}$, $\mathbb{P}(A \cap B) = 0.3$

Because $A \cap C = \emptyset$, $\mathbb{P}(A \cap C) = 0$

Example 4

In a community, 32% of the population are male smokers; 27% are female smokers. What percentage of the population of this community smoke?

Solution Let A be the event that a randomly selected person from this community smokes. Let B be the event that the person is male. By equation (11)

$$\mathbb{P}(A) = \mathbb{P}(A \cap B) + \mathbb{P}(A \cap B^c) = 0.32 + 0.27 = 0.59$$

Therefore 59% of the community smokes.

Example 5

Find example in tutorial set 1

Solution to Example 5

Solution

Let E, F, and G be the events that the person reads A, B, and C, respectively. The event that the person reads at least one of the newspapers A, B, or C is $E \cup F \cup G$. Therefore, $1 - P(E \cup F \cup G)$ is the probability that he or she reads none of them. Since

$$\begin{aligned}\mathbb{P}(E \cup F \cup G) &= \mathbb{P}(E) + \mathbb{P}(F) + \mathbb{P}(G) - \mathbb{P}(E \cap F) - \mathbb{P}(E \cap G) \\ &\quad - \mathbb{P}(F \cap G) + \mathbb{P}(E \cap F \cap G) \\ &= 0.25 + 0.20 + 0.13 - 0.10 - 0.08 - 0.05 + 0.04 \\ &= 0.39\end{aligned}$$

The desired probability equals 1-0.39 = 0.61

Exercise

Question Dr. Grossman, an internist, has 520 patients, of which 230 are hypertensive, 185 are diabetic, 35 are hypochondriac and diabetic, 25 are all three, 150 are none, 140 are only hypertensive, and finally, 15 are hypertensive and hypochondriac but not diabetic. Find the probability that Dr. Grossman's next appointment is hypochondriac but neither diabetic nor hypertensive. Assume that appointments are all random. This implies that even hypochondriacs do not make more visits than others.

Answer $= \frac{30}{520} \approx 0.06$

I-4-3: Independent Events

Definition (Independent Events)

Two events A and B are said to be independent if

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B) \tag{12}$$

If two events are not independent, they are called **dependent**. If A and B are independent, we say that $\{A, B\}$ is an independent set of events.

Theorem

If A and B are independent, then A and B^c are independent as well.

$$\mathbb{P}(A \cap B^c) = \mathbb{P}(A)\mathbb{P}(B^c) \tag{13}$$

I-4-3: Continued

If A and B are independent, then A^c and B^c are independent as well.

$$\mathbb{P}(A^c \cap B^c) = \mathbb{P}(A^c)\mathbb{P}(B^c) \tag{14}$$

Definition (Independent Events)

The events A, B, and C are called independent if

$$\begin{aligned} P(A \cap B) &= P(A)P(B), \\ P(A \cap C) &= P(A)P(C), \\ P(B \cap C) &= P(B)P(C), \\ P(A \cap B \cap C) &= P(A)P(B)P(C). \end{aligned}$$

If A, B, and C are independent events, we say $\{A, B, C\}$ is an independent set of events.

Example 1

Assume that the probability that a wafer (a thin slice of semiconductor) contains a large particle of contamination is 0.01 and that the wafers are independent; that is, the probability that a wafer contains a large particle is not dependent on the characteristics of any of the other wafers. If 15 wafers are analyzed, what is the probability that no large particles are found?

Solution

Let E_i denote the event that the *ith* wafer contains no large particles, Then, $i = 1, 2, \cdots, 15$ and $\mathbb{P}(E_i) = 0.99$.
From the independence assumption

$$\begin{aligned}\mathbb{P}(E_1 \cap E_2 \cap \cdots \cap E_{15}) &= \mathbb{P}(E_1) \times \mathbb{P}(E_2) \times \cdots \times \mathbb{P}(E_{15}) \\ &= 0.99^{15} = 0.89\end{aligned}$$

Example 2

The circuit of Figure (1) operates only if there is a path of functional devices from left to right. The probability that each device functions is shown . Assume that devices fail independently. What is the probability that the circuit operates?

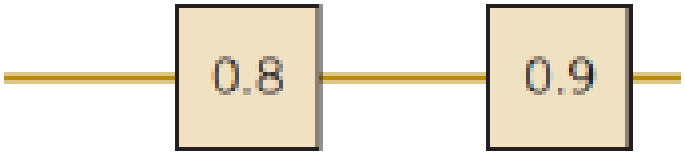


Figure 1: Series Circuit

Solution

Let L and R denote the events that the left and right devices operate, respectively. There is only a path if both operate. From the independence assumption

$$\mathbb{P}(L \cap R) = \mathbb{P}(L) \times \mathbb{P}(R) = 0.8 \times 0.9 = 0.72$$

Example 3

Let an experiment consist of throwing a die twice. Let A be the event that in the second throw the die lands 1, 2, or 5; B the event that in the second throw it lands 4, 5 or 6; and C the event that the sum of the two outcomes is 9. Then
$P(A) = P(B) = 1/2, P(C) = 1/9$ and

$$\mathbb{P}(A \cap B) = \frac{1}{6} \neq \frac{1}{4} = \mathbb{P}(A)\mathbb{P}(B)$$
$$\mathbb{P}(A \cap C) = \frac{1}{36} \neq \frac{1}{18} = \mathbb{P}(A)\mathbb{P}(C)$$
$$\mathbb{P}(B \cap C) = \frac{1}{12} \neq \frac{1}{18} = \mathbb{P}(B)\mathbb{P}(C)$$

But $\mathbb{P}(A \cap B \cap C) = \mathbb{P}(A)\mathbb{P}(B)\mathbb{P}(C) = \frac{1}{36}$, though $\mathbb{P}(A \cap B \cap C) = \mathbb{P}(A)\mathbb{P}(B)\mathbb{P}(C)$, this is not sufficient for the independence of A, B, C

Exercise

Find applications and more examples in tutorial set 1

I-4-4: Conditional Probability

Definition

Let A and B be two events in the sample space, Ω with $\mathbb{P}(B) > 0$, the **conditional probability** of A given B, denoted by $\mathbb{P}(A \mid B)$, is

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)} \tag{15}$$

If A and B are independent then

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)} = \frac{\mathbb{P}(A)\mathbb{P}(B)}{\mathbb{P}(B)} = \mathbb{P}(A)$$

and

$$\mathbb{P}(B \mid A) = \mathbb{P}(B)$$

Example

A family has two children. What is the conditional probability that both are boys given that at least one of them is a boy? Assume that the sample space S is given by

$$S = \{(b, b), (b, g), (g, b), (g, g)\}$$

and all outcomes are equally likely.

Solution

Let B denote the event that both children are boys, and A the event that at least one of them is a boy, then the probability we want to find is given by

$$\mathbb{P}(B \mid A) = \frac{\mathbb{P}(B \cap A)}{\mathbb{P}(A)} = \frac{\mathbb{P}(b, b)}{\mathbb{P}((b, b), (b, g), (g, b))} = \frac{1/4}{3/4} = \frac{1}{3}$$

Example

Find example in tutorial set 1

Example

A day's production of 850 manufactured parts contains 50 parts that do not meet customer requirements. Two parts are selected without replacement from the batch. What is the probability that the second part is defective given that the first part is defective.

Solution

Let A denote the event that the first part selected is defective, and let B denote the event that the second part selected is defective. The probability needed can be expressed as $\mathbb{P}(B \mid A)$,

$$\mathbb{P}(B \mid A) = \frac{49}{849}$$

If 3 parts are selected at random, what is the probability that the first two are defective and the third one is not. Ans $= 0.0032$

Example

Consider the experiment of rolling a coin, $\Omega = \{1, 2, 3, 4, 5, 6\}$ Let the event $A = \{1, 2, 3\}$, $\mathbb{P}(A) = \frac{1}{2}$ and $B = \{3, 4\}$, $\mathbb{P}(B) = \frac{1}{3}$. $A \cap B = \{3\}$, $\mathbb{P}(A \cap B) = \frac{1}{6}$ Are A and B independent?

$$\mathbb{P}(A \mid B) = \frac{1}{2} = \mathbb{P}(A) \text{ and } \mathbb{P}(B \mid A) = \frac{1}{3} = \mathbb{P}(B)$$

Therefore A and B are independent. If one holds the other will hold

I-4-4 Continued : The Law of Multiplication

The Law of multiplication calculates the probabilities of the intersection of events in terms of conditional probabilities of later events given earlier events.

Equation (15) is useful for calculating $\mathbb{P}(A \cap B)$. Multiplying both sides of equation (15) by $\mathbb{P}(B)$ we obtain

The Law of Multiplication

$$\mathbb{P}(A \cap B) = \mathbb{P}(B) \times \mathbb{P}(A \mid B), \;\; \mathbb{P}(B) > 0$$

which means that the probability of the joint occurrence of A and B is the product of the probability of B and the conditional probability of A given that B has occurred. Similarly;

I-4-4: Continued

The Law of Multiplication

$$\mathbb{P}(A \cap B) = \mathbb{P}(A) \times \mathbb{P}(B \mid A), \;\; \mathbb{P}(A) > 0 \tag{16}$$

Question: Suppose that five good fuses and two defective ones have been mixed up. To find the defective fuses, we test them one-by-one, at random and without replacement. What is the probability that we are lucky and find both of the defective fuses in the first two tests?

Solution

Let A_1 and A_2 be the events of finding a defective fuse in the first and second tests, respectively. We are interested in $\mathbb{P}(A_1 \cap A_2)$. From equation (16) we have

$$\mathbb{P}(A_1 \cap A_2) = \mathbb{P}(A_1) \times \mathbb{P}(A_2 \mid A_1) = \frac{2}{7} \times \frac{1}{6} = \frac{1}{21} \tag{17}$$

Example

The probability that the first stage of a numerically controlled machining operation for high-rpm pistons meets specifications is 0.90. Given that the first stage meets specifications, the probability that a second stage of machining meets specifications is 0.95. What is the probability that both stages meet specifications?

Solution

Let A and B denote the events that the first and second stages meet specifications, respectively. The probability that

$$\mathbb{P}(A \cap B) = \mathbb{P}(A) \times \mathbb{P}(B \mid A) = 0.9 \times 0.95 = 0.855$$

Example

Suppose a box contains 20 items of which 9 are defective. Two items are drawn at random from from the batch one after the other, without replacement. What is the probability that
(a) both are defective?
(b) the second item is defective?

Solution

Let D_1 and D_2 be events that first and second item drawn is defective respectively

$$\mathbb{P}(D_1) = \frac{9}{20}, \quad \mathbb{P}(D_2 \mid D_1) = \frac{8}{19}, \quad \mathbb{P}(D_2 \mid D_1^c) = \frac{9}{19}$$

Solution to Example

(a) The probability that both will be effective

$$\begin{aligned}\mathbb{P}(D_1 \cap D_2) &= \mathbb{P}(D_1)\mathbb{P}(D_2 \mid D_1) \\ &= \frac{9}{20} \times \frac{8}{19} = \frac{18}{95}\end{aligned}$$

(b) The probability that the second is defective is

$$\begin{aligned}\mathbb{P}(D_1 D_2 \text{ or } D_1^c D_2) &= \mathbb{P}(D_1 \cap D_2) + \mathbb{P}(D_1^c \cap D_2) \\ &= \mathbb{P}(D_1)\mathbb{P}(D_2 \mid D_1) + \mathbb{P}(D_1^c)\mathbb{P}(D_2 \mid D_1^c) \\ &= \left(\frac{9}{20} \times \frac{8}{19}\right) + \left(\frac{11}{20} \times \frac{9}{19}\right) = \frac{9}{20}\end{aligned}$$

Exercise

1 Let A and B be events such that $\mathbb{P}(A) = 0.6$ and $\mathbb{P}(B) = 0.5$ and $\mathbb{P}(A \cup B) = 0.8$. Find $\mathbb{P}(A \mid B)$. Are A and B independent?

The Law of Total Probability

It is not always possible to calculate directly the probability of the occurrence of an event $A \in \Omega$, but it is possible to find $\mathbb{P}(A \mid B)$ and $\mathbb{P}(A \mid B^c)$

The Law of total probability calculates the probabilities of unconditional events in terms of conditional probabilities of later events given the earlier ones.

I-4-4 : The Total Law of Probability

Theorem (The Law of Total Probability)

Let B be an event with $\mathbb{P}(B) > 0$ and $\mathbb{P}(B^c) > 0$. The for any event A of Ω,

$$\mathbb{P}(A) = \mathbb{P}(A \mid B)\mathbb{P}(B) + \mathbb{P}(A \mid B^c)\mathbb{P}(B^c)$$

It states that $\mathbb{P}(A)$ is the weighted average of the probability of A given that B has occurred and probability of A given that B has not occurred.

Example

A manufacturing company rents 35% of the cars for its customers from agency *I* and 65% from agency *II*. If 8% of the cars of agency *I* and 5% of the cars of agency *II* break down during the rental periods, what is the probability that a car rented by this manufacturing company breaks down?

Solution

Let A be the event that a car rented by this company breaks down. Let I and II be the events that it is rented from agencies I and II, respectively. Then by the law of total probability,

$$\begin{aligned}\mathbb{P}(A) &= \mathbb{P}(A \mid I)\mathbb{P}(I) + \mathbb{P}(A \mid II)\mathbb{P}(II) \\ &= (0.08)(0.35) + (0.05)(0.65) = 0.0605.\end{aligned}$$

Tree Representation

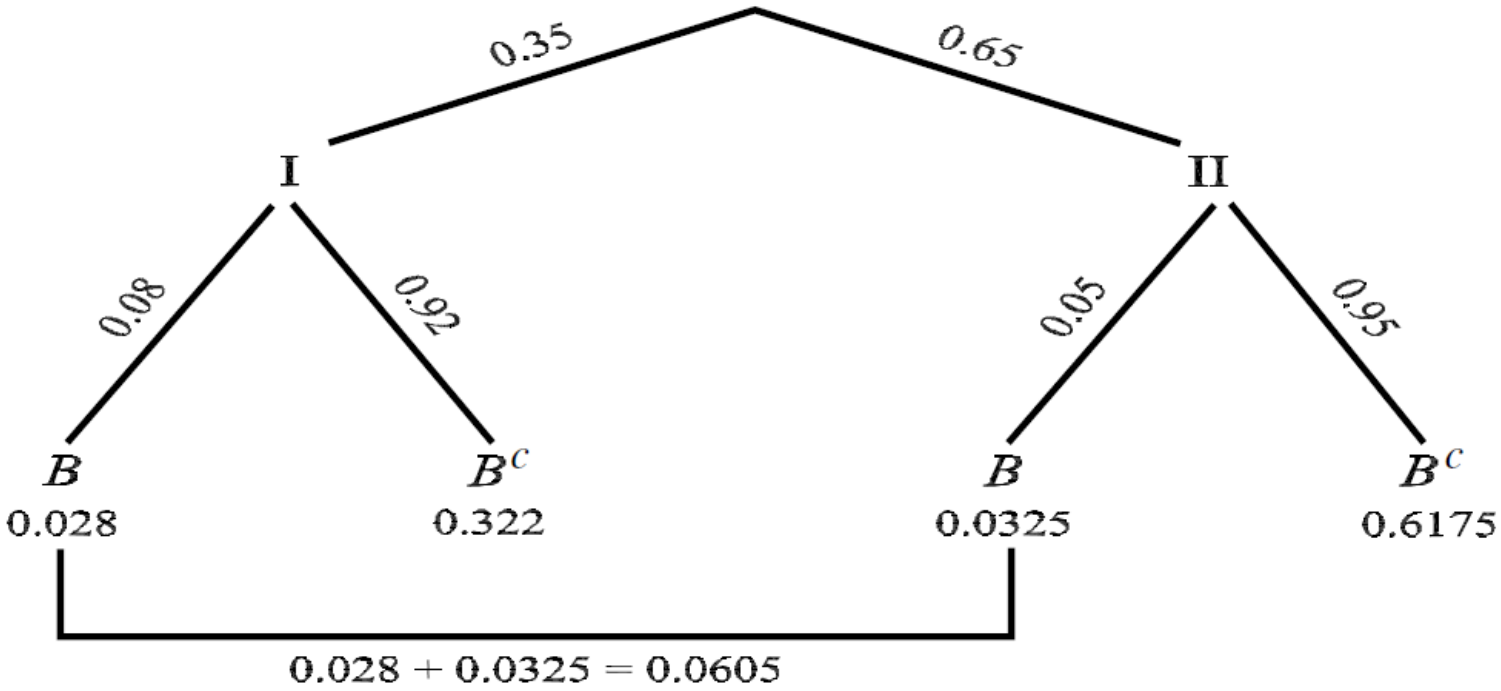


Figure 2: Tree representation of the example above

Law of Total Probability

The Law of Total Probability

Suppose $B_1, B_2, B_3, \cdots B_n$ are disjoint events such that $B_1 \cup B_2 \cup B_3 \cup \cdots \cup B_n = \Omega$ and $\mathbb{P}(B_i) > 0$ for $i = 1, 2, \cdots, n$, then the probability of an arbitrary event $A \in \Omega$ can be expressed as

$$\begin{aligned}\mathbb{P}(A) &= \mathbb{P}(A \mid B_1)\mathbb{P}(B_1) + \mathbb{P}(A \mid B_2)\mathbb{P}(B_2) + \cdot + \mathbb{P}(A \mid B_n)\mathbb{P}(B_n) \\ &= \sum_{i=1}^{n} \mathbb{P}(A \mid B_i)\mathbb{P}(B_i)\end{aligned}$$

Example

In a certain assembly plants, three machines A, B and C make 60%; 25% and 15%, respectively, of the products. It is known from past experience that 6% of the products made by machine A, 4% of the products made by machine B and 2% of the products made by machine C are defective. If a finished product is selected at random, what is the probability that it is defective.

Solution

Let A_1, A_2, A_3 denote the events that the finished product was made by machine A, B and C respectively. Let D denote the event that the finished product is defective. Our task is to find P[D]. We have

$$\mathbb{P}(A_1) = 0.6; \mathbb{P}(A_2) = 0.25; \mathbb{P}(A_3) = 0.15; \mathbb{P}(D \mid A_1) = 0.06;$$
$$\mathbb{P}(D \mid A_2) = 0.04; \mathbb{P}(D \mid A_3) = 0.02$$

Sketch the tree diagram representation

Example Continued

A_1, A_2 and A_3 form a partition of the sample space. Hence

$$\begin{aligned}\mathbb{P}(D) &= \mathbb{P}(A_1)\mathbb{P}(D \mid A_1) + \mathbb{P}(A_2)\mathbb{P}(D \mid A_2) + \mathbb{P}(A_3)\mathbb{P}(D \mid A_3) \\ &= 0.6(0.06) + 0.25(0.04) + 0.15(0.02) \\ &= 0.049\end{aligned}$$

Question α: In a semiconductor manufacturing company, assume the following probabilities for product failure subject to levels of contamination in manufacturing

Probability of Failure	Level of Contamination
0.1	High
0.01	Medium
0.001	Low

Example Continued

In one of their production runs, 20% of the chips are subjected to high levels of contamination, 30% to medium levels of contamination, and 50% to low levels of contamination. What is the probability that a product using one of these chips fails?

Solution Let H denote the event that a chip is exposed to high levels of contamination, M denote the event that a chip is exposed to medium levels of contamination, L denote the event that a chip is exposed to low levels of contamination Then,

$$\begin{aligned}\mathbb{P}(F) &= \mathbb{P}(H)\mathbb{P}(F \mid H) + \mathbb{P}(M)\mathbb{P}(F \mid M) + \mathbb{P}(L)\mathbb{P}(F \mid L) \\ &= 0.10(0.20) + 0.01(0.30) + 0.001(0.50) \\ &= 0.0235\end{aligned}$$

Tree Diagram

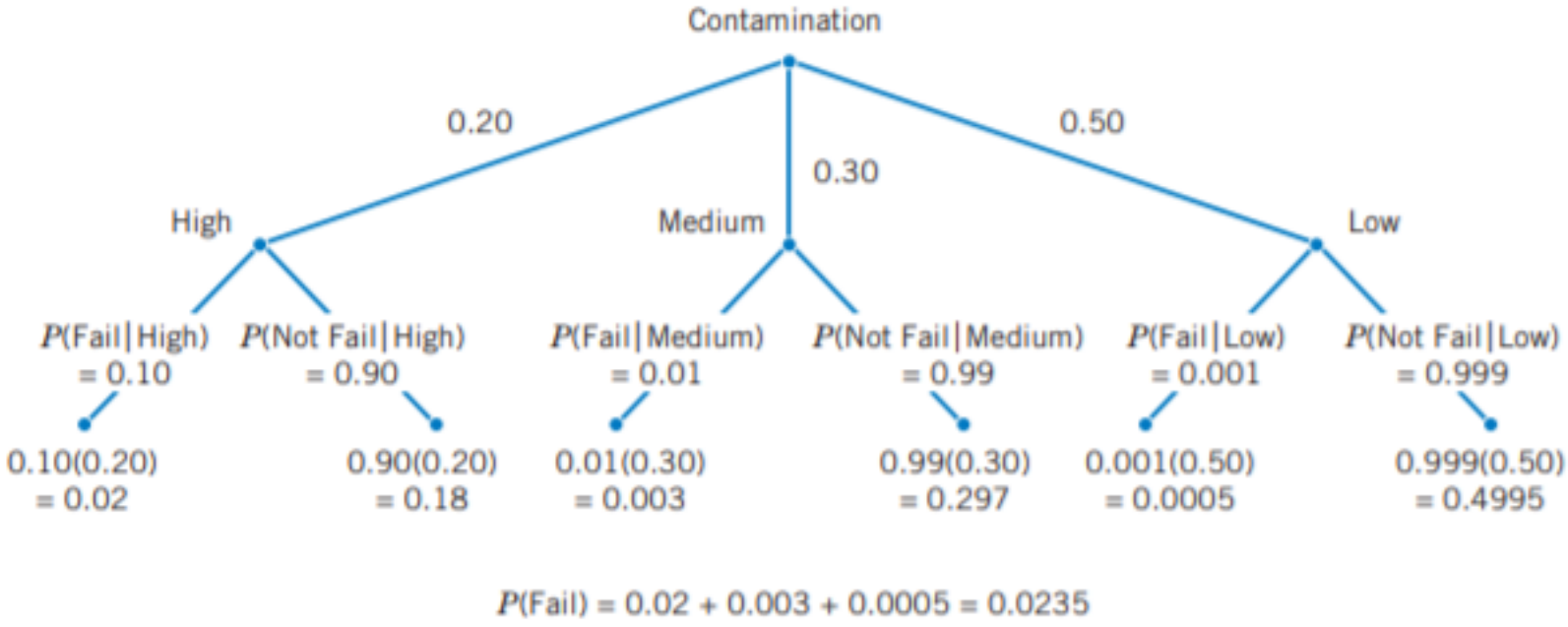


Figure 3: Tree Diagram Representation

A tree diagram is helpful in arriving at the required probability

Exercises

1. A population is composed of 55% men and 45% women. It is known that 80% of the men and 15% of the women smoke cigarettes. What is the probability that a person selected at random from this population smokes?
2. An insurer notes that 50% of its policyholders have health coverage. 60% of policyholders with health coverage also have dental coverage. 15% of the policyholders have dental insurance and not health insurance. Find the probability that a randomly selected policyholder has dental insurance.

Bayes' Theorem

Let $\{B_1, B_2, \cdots B_n\}$ be a partition of the sample space, Ω of an experiment. If for $i = \{1, 2, \cdots, n\}$, $\mathbb{P}(B_i) > 0$, then for any arbitrary event A of Ω with $\mathbb{P}(A) > 0$ the conditional probability of B_i, given A is

$$\mathbb{P}(B_k \mid A) = \frac{\mathbb{P}(A \cap B_k)}{\mathbb{P}(A)}$$
$$= \frac{\mathbb{P}(A \mid B_k)\mathbb{P}(B_k)}{\mathbb{P}(A \mid B_1)\mathbb{P}(B_1) + \mathbb{P}(A \mid B_2)\mathbb{P}(B_2) + \cdots + \mathbb{P}(A \mid B_n)\mathbb{P}(B_n)} \quad (18)$$

In statistical application of Bayes' theorem $B_1, B_2, \cdots B_n$ are called hypothesis, $\mathbb{P}(B_i)$ is called the prior probability of B_i and the conditional probability of $\mathbb{P}(B_i \mid A)$ is called the posterior probability of B_i after the occurrence of A.

Bayes' Theorem

Note that for any event B of Ω, B and B^c both nonempty, the set $\{B, B^c\}$ is a partition of Ω. Thus by the Bayes' theorem. If $\mathbb{P}(B) > 0$ and $\mathbb{P}(B^c) > 0$ then for any event A of Ω with $\mathbb{P}(A) > 0$

$$\mathbb{P}(B \mid A) = \frac{\mathbb{P}(A \mid B)\mathbb{P}(B)}{\mathbb{P}(A \mid B)\mathbb{P}(B) + \mathbb{P}(A \mid B^c)\mathbb{P}(B^c)} \tag{19}$$

Similarly,

$$\mathbb{P}(B^c \mid A) = \frac{\mathbb{P}(A \mid B^c)\mathbb{P}(B^c)}{\mathbb{P}(A \mid B)\mathbb{P}(B) + \mathbb{P}(A \mid B^c)\mathbb{P}(B^c)} \tag{20}$$

Equations (19) and (20) are the simplest forms of the Bayes' formula.

Example

Reconsider Question α with values given in Table (1). The conditional probability that a high level of contamination was present when a failure occurred is to be determined.

Probability of Failure	Level of Contamination	Probability of Level
0.1	High	0.2
0.005	Not High	0.8

Table 1: Contamination

We want to find $\mathbb{P}(H \mid F)$ which is given as

$$\mathbb{P}(H \mid F) = \frac{\mathbb{P}(F \mid H)\mathbb{P}(H)}{\mathbb{P}(F \mid H)\mathbb{P}(H) + \mathbb{P}(F \mid H^c)\mathbb{P}(H^c)} = \frac{(0.10)(0.20)}{0.024} = 0.83$$

Example

In a certain assembly plant, three machines A, B,C make 30%, 45%, 25% respectively of the products. It is known from past experience that 2% of the product made by machine A, 3% of machine B and 2% of machine C are defective. Given that a product is defective, what is the probability that it is produced by machine A?

Solution: We want to find $\mathbb{P}(A \mid D)$

$$\mathbb{P}(A) = 0.30; \mathbb{P}(B) = 0.45; \mathbb{P}(C) = 0.25$$
$$\mathbb{P}(D \mid A) = 0.02; \mathbb{P}(D \mid B) = 0.03; \mathbb{P}(D \mid C) = 0.02$$

$$\begin{aligned}\mathbb{P}(D) &= \mathbb{P}(A)\mathbb{P}(D \mid A) + \mathbb{P}(B)\mathbb{P}(D \mid B) + \mathbb{P}(C)\mathbb{P}(D \mid C) \\ &= 0.02(0.3) + 0.03(0.45) + 0.02(0.25) \\ &= 0.0245\end{aligned}$$

Example Continued

From equation (18)

$$\mathbb{P}(A \mid D) = \frac{\mathbb{P}(D \mid A)\mathbb{P}(A)}{\mathbb{P}(D)} = \frac{(0.02)(0.3)}{0.0245} = 0.245$$

Question At a certain stage of a criminal investigation, the inspector in charge is 60 percent convinced of the guilt of a certain suspect. Suppose, however, that a new piece of evidence which shows that the criminal has a certain characteristic (such as left-handedness, baldness, or brown hair) is uncovered. If 20 percent of the population possesses this characteristic, how certain of the guilt of the suspect should the inspector now be if it turns out that the suspect has the characteristic?

Example Continued

Let G denote the event that the suspect is guilty and C the event that he possesses the characteristic of the criminal, we have

$$\begin{aligned}\mathbb{P}(G \mid C) &= \frac{\mathbb{P}(G \cap C)}{\mathbb{P}(C)} \\ &= \frac{\mathbb{P}(C \mid G)\mathbb{P}(G)}{\mathbb{P}(C \mid G)\mathbb{P}(G) + \mathbb{P}(C \mid G^c)\mathbb{P}(G^c)} \\ &= \frac{(1)(0.60)}{(1)(0.60) + 0.2(0.4)} = 0.882\end{aligned}$$

NB: We supposed that the probability of the suspect having the characteristic if he is, in fact, innocent is equal to .2, which is also the proportion of the population possessing the characteristic

Exercise:Bayes' Network

Bayesian networks are used on the Web sites of high technology manufacturers to allow customers to quickly diagnose problems with products. A printer manufacturer obtained the following probabilities from a database of test results. Printer failures are associated with three types of problems: hardware, software, and other (such as connectors), with probabilities 0.1, 0.6, and 0.3, respectively. The probability of a printer failure given a hardware problem is 0.9, given a software problem is 0.2, and given any other type of problem is 0.5. If a customer enters the manufacturer's Web site to diagnose a printer failure, what is the most likely cause of the problem?

NB: The most likely cause of the problem is the one that corresponds to the largest probability.

Thank you

For the second part of this course we will look at Random Variables and Probability Distributions. All the Best and Keep Reading.

THANK YOU